



College of Engineering
ELEC 491 – Electrical Engineering Design Project
Final Report

AUTOMATIC LIQUID MIXER ROBOT

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Abstract	3
1. Introduction	4
2. System Design	5
2.1 Overview	5
2.2 Liquid Dispenser	6
2.2.1 Overview	6
2.2.2 Algorithm used for Huzzah	7
2.2.3 Algorithm used for Mega	7
2.3 Cart Mechanism	7
2.3.1 Overview	7
2.3.2 Algorithm of Arduino Micro	8
2.3.3 Distance Sensors Calculation	8
2.4 Mobile Application	9
2.5 Communication Between Application and the Liquid Dispenser Mechanism	10
2.6 Communication between Cart and Liquid Dispenser	10
2.7 Chassis	11
3. Analysis and Results	11
3.1 Overview	11
3.2 Results	12
3.3 Limitations	12
4. Conclusions	12
5. References	13
6. Appendixes	13

Abstract

We designed and built an IoT based Robot that is able to prepare mixtures by receiving orders from a mobile application. The application can be used on both IOS and Android, since it was developed using cross-platform tools. This project can be utilised in cafes and restaurants or in households. It has many potential and it can improve in many different directions to be utilised in further ways. The design consists of 3 main components: cart mechanism, liquid dispenser mechanism and the mobile application. User places the glass on the cart and orders from the application. Cart prepares the mixture by going to each pump one by one. The cart and liquid dispenser communicates and the liquid amount from each pump is dispensed at the right time in the right amount. Then the user can take the glass as it will be prepared after the cart finishes travelling to each pump one by one.

1. Introduction

The objective of this project was to create a robot-application pair that would create mixtures of liquids as the user orders them. The application and the robot are both connected to the wifi through which they communicate the user's order. The user places a glass on the cart and the cart moves along one axis to make the glass travel to each pump one by one. The cart and the liquid dispenser communicate via bluetooth. The liquid dispenser has the 'brains' of the robot which coordinates the preparation of the mixture.

This project can be used in commercial use. It can be used in cafes and restaurants to prepare drinks in an entertaining manner. Or it can be used as a household appliance where it can be used for many different purposes. It can be integrated with other fields such as health to prepare custom mixtures specific to that field. For example in the field of health it can prepare a mixture that lowers blood pressure automatically if the user's blood pressure is measured to be higher than normal.

The application allows the user to express the wanted mixture in a variety of ways. The user can directly enter the amount of liquid or set the maximum amount of mixture and then choose percentages of each liquid. The user can also use preset mixtures to order a mixture. These appear when the liquids for it are added to the pumps and the user can select which liquid is added to which pump.

There were several similar projects that we discovered while searching the internet. Our project is different from the others in various ways. One of the similar projects we have found [1] does not utilise a cart mechanism but instead, connects all the straws to one output and manually places the cup underneath the connected straws. Another example [2] utilises a mobilised cart system but their liquid delivery system is based upon a manual press and hold mechanism where the bottles are upside down and thus using gravitational force to pour the liquids into the cup and the recipe ordering is a non-IoT system directly connected to the controller. Another example [3] we have found on the internet was a company building end-products very similar to our project. However this product's design is not accessible to the public and it appears that it doesn't use a pump based mechanism due to the fact that the bottles are upside down. Which led us to believe that this system uses the gravitational force as the delivery method as well. Another difference between this product and our project is that the product in question utilises a bluetooth based system to connect to the user whereas we used a WiFi connection.

Our project can be summarised as follows:

1. Utilises IoT to receive order from user via a mobile application
2. Has a cart system to move the glass from pump to pump
3. Uses pumps to serve 4 distinct liquids
4. Utilises 3 microcontrollers to deliver the correct mixture to the user.

2. System Design

2.1 Overview

The overview of our project can be seen in figure 1. It composes of three parts: The mobile application, the liquid dispenser system, and the cart system. The latter two form the robot. The user interacts with both the robot and the mobile application. User places a glass on top of the cart mechanism on the cup holder and uses the application to make an order. The application sends the order via WIFI to a remote server from which the robot reads the order and starts preparing the mixture. After the robot is done, the user can take the glass. Inside the robot cart and dispenser mechanism coordinate via communication with bluetooth.

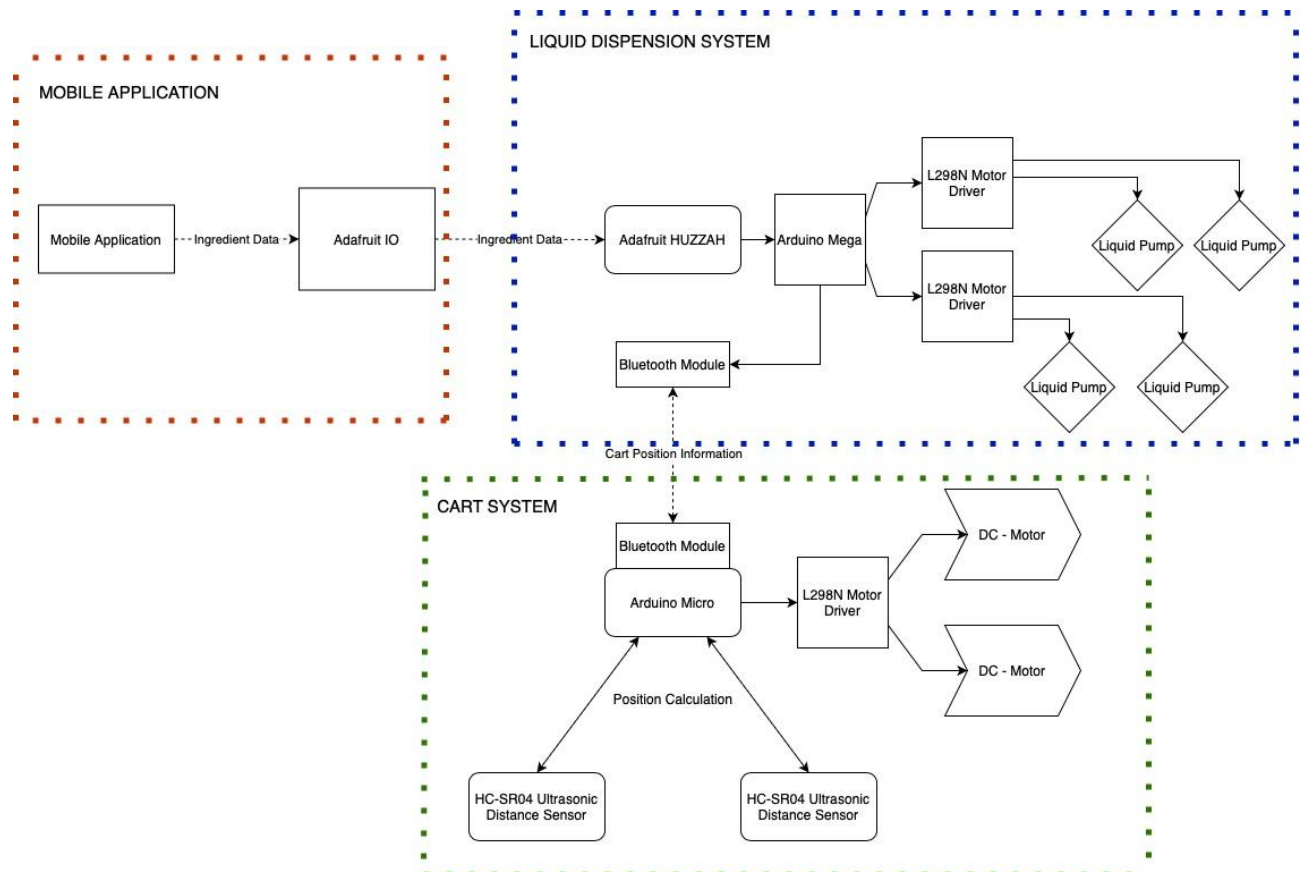


Figure 1: Overview of the Project

2.2 Liquid Dispenser

2.2.1 Overview

The liquid dispenser mechanism consists of 4 pumps, 2 motor drivers, 1 arduino mega, 1 adafruit huzzah [4], and 1 bluetooth module. The pumps are connected to drivers, the drivers and the bluetooth module are connected to arduino mega and the huzzah has several connections allowing bidirectional communication with arduino mega as will be described in the following sections.

We experimented with the pumps and learned that each pump dispenses 60-70 millilitres of liquid in 1 minute. When we divide it by 60, we get around 1 millilitre per second. This way we can translate the user's requested amount to the time of pumps running.

The huzzah and the mega are connected via 5 cables in total. One of the cables is for the huzzah to receive feedback from mega whereas the other 4 are for the huzzah to tell which pump the cart should go to and which liquid should be dispensed. Thus 4 of them provide communication from huzzah to mega and one provides the reverse way. Since mega works with 5V logic and huzzah works with 3.3V we added a voltage divider circuit which drops the 5V to approximately 3.4V.

The mega is powered by a 9V battery. The motor driver is powered by the power supply's 12V output. The huzzah is powered by the mega's 5V output.

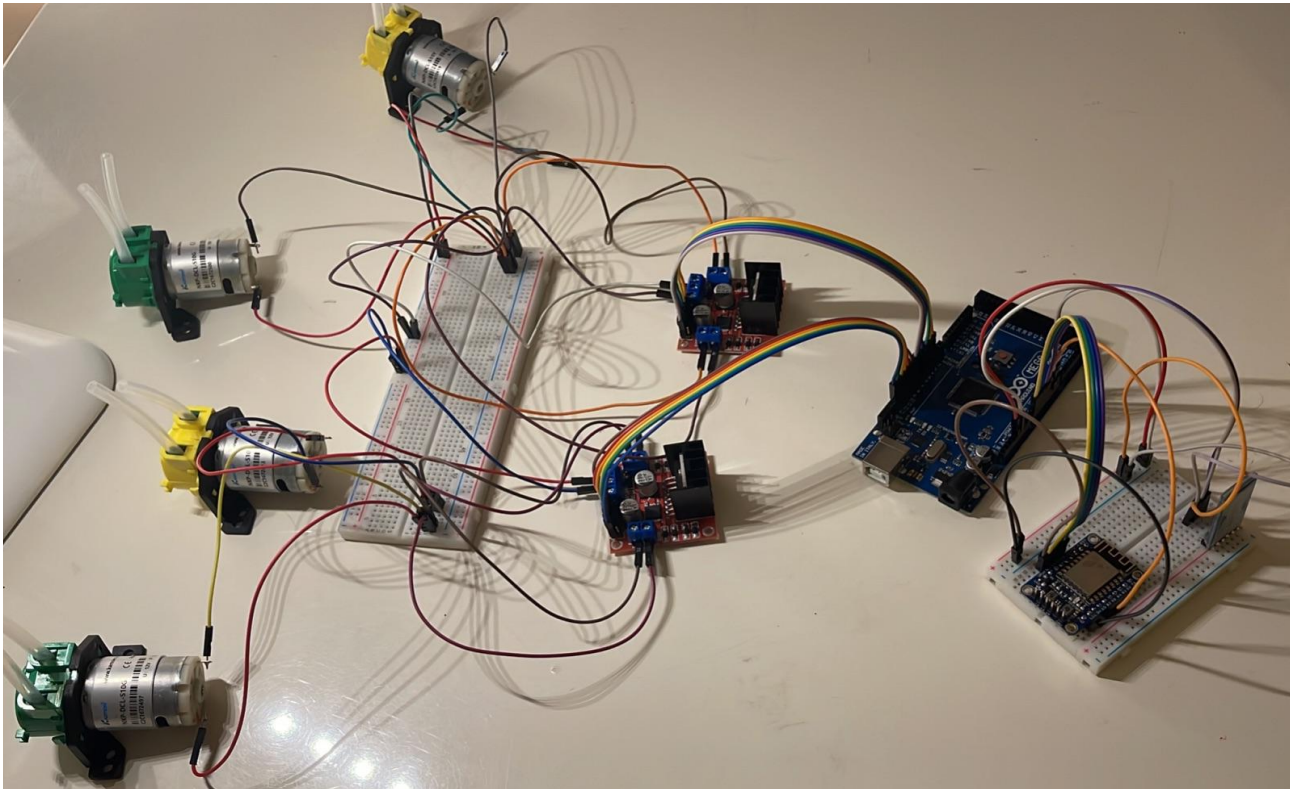


Figure 2: Liquid Dispenser Mechanism

2.2.2 Algorithm used for Huzzah

The huzzah waits for a message to arrive at the server. After receiving the message, it sends the mega first signal (which pump cart should go to). After receiving confirmation from mega that the cart has gone to the pump it tells mega to start the pump and then tells it to stop the pump. Since it uses 3.3V logic, the mega is tuned to detect voltages according to this level by doing analog reads. It continues this for each liquid included.

2.2.3 Algorithm used for Mega

The mega waits for the huzzah to send it information. When it receives the pump number it sends this to the cart via bluetooth and waits for it to come to the right position. Upon receiving the correct position it sends the huzzah the confirmation and huzzah tells it to start pouring. The mega then starts the correct pump till huzzah tells it to stop. Then this continues for each liquid included.

2.3 Cart Mechanism

2.3.1 Overview

The cart mechanism, which can be seen in figure 3, is composed of 1 arduino micro, 4 motors, 1 motor driver, 2 distance sensors, and 1 bluetooth. The motors are connected to the motor driver,

front motors work together and the back motors work together. Everything else is connected to the arduino micro.

The Arduino Micro is powered via a 9V battery, whereas the motor driver is powered by the power supply's 12V output.

2.3.2 Algorithm of Arduino Micro

The brief explanation of the algorithm is as follows: The cart waits for the mega to send it the pump address to go to. Then it reads distance measurements and gets an estimate of under which pump it is (this estimate could be 2.0 for example meaning it is under pump #2, or 2.5 meaning it is between pumps #2 and #3). Then it moves in the direction it needs to according to the pump it is going to and the location it is at. After it goes to a pump it sends its location to the liquid system and waits for the next location to go to. This repeats endlessly.

2.3.3 Distance Sensors Calculation

We utilised 2 distance sensors. These sensors made the cart capable of estimating at which pump it currently is. This meant that it knew which way to go and when to stop. We did this by utilising two distance sensors because we saw that the distance sensor did not work very well when the distance was small. So, we used two distance measurements, one for the front and one for the back. The arduino micro took both of these measurements and used the measurement that was largest and also in a meaningful range. We saw that too close readings led to unmeaningful readings such as >1000 cm readings, thus we discarded such a reading by looking at the opposite sensor. Also, by discarding the smallest reading we got to use the largest meaningful reading. We saw that our robot was capable of predicting its location quite accurately.

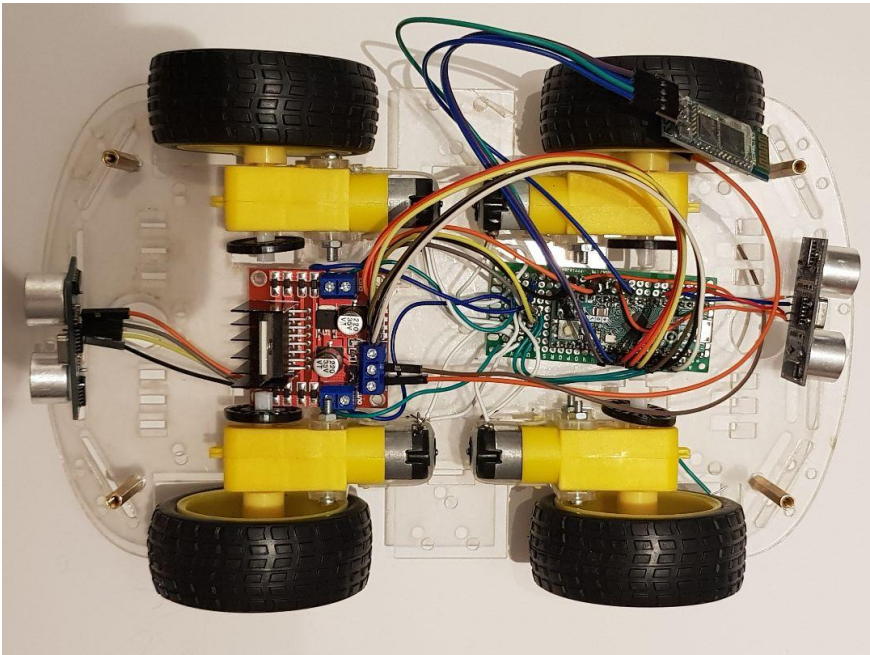


Figure 3: Cart Mechanism

2.4 Mobile Application

The application was built using dart and flutter [5]. This allowed us to build it in one code base only and then use it on both IOS and Android. The application's homepage is shown on figure 4 where users can choose drink size, and drink names. Users can also enter new drink names. On figure 5 we can see the custom mixture selection screen. This screen allows the user to specify amounts of each liquid and make such an order. Alternatively users can select percentage to make the mixture percentage-wise by specifying parts. A second option is make preset mixture which shows some preset mixtures that are makeable with the given drinks list.

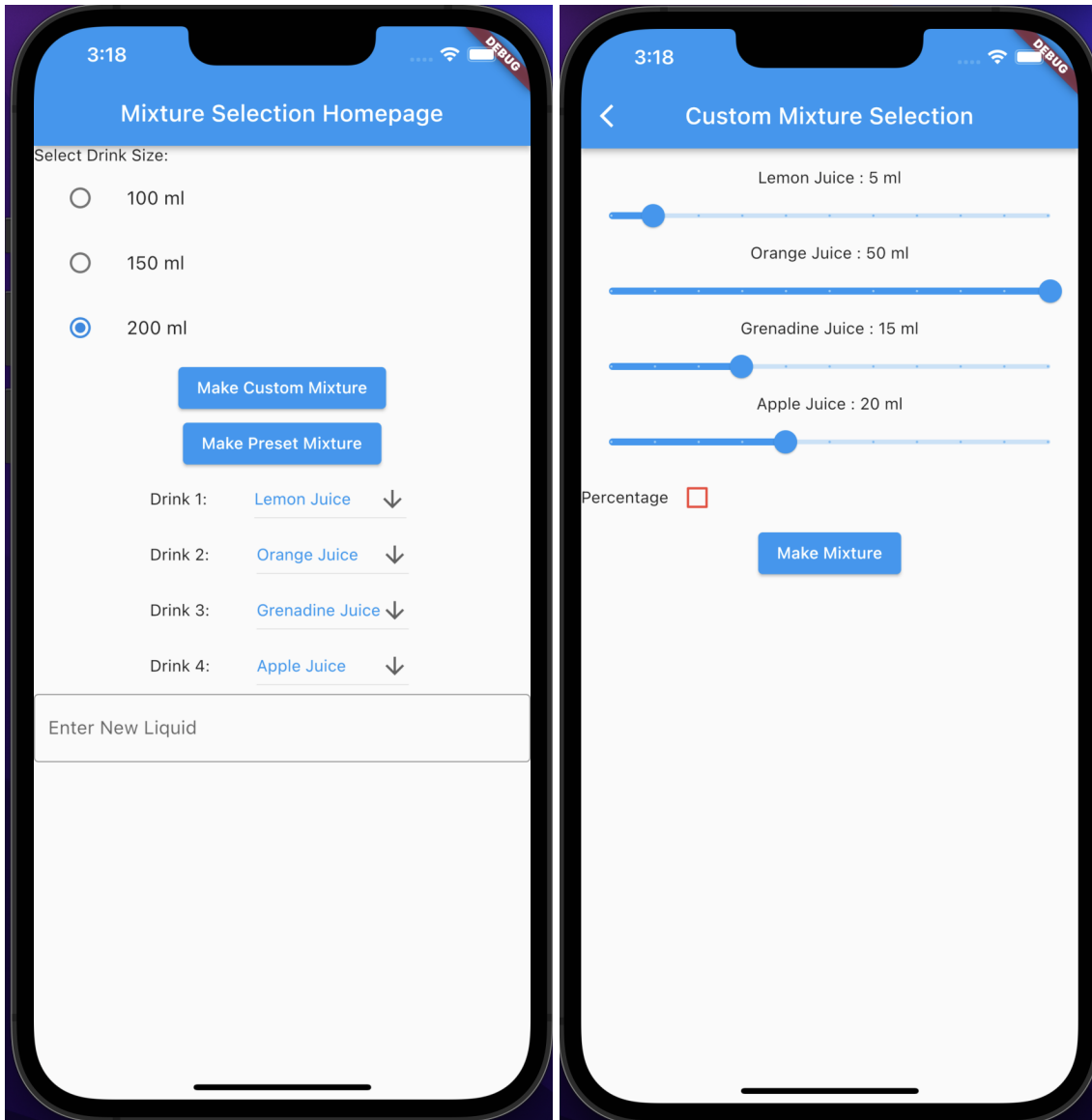


Figure 4 and 5: Application homepage and custom mixture page

2.5 Communication Between Application and the Liquid Dispenser Mechanism

The application and the Huzzah in the liquid dispenser mechanism communicate via wifi. The application uses a key belonging to the user for a server on adafruit io and sends a http request to this server with the order. The huzzah in the robot reads this from the server and starts preparing the mixture immediately. We saw that the wifi worked very fast despite it having to travel to a server.

2.6 Communication between Cart and Liquid Dispenser

The cart and the liquid dispenser communicate bidirectional by the use of bluetooth. The cart sends its location every time it is at a pump's vicinity and the mega sends the information of which pump cart needs to go to.

We saw that at least one of these bluetooth modules needs to be capable of being in the role ‘master’ (or 1) which only some bluetooth modules can do. The two modules when linked enable effortless communication and they work robustly. We saw that they worked even when the battery levels were low.

2.7 Chassis

We made the chassis following the measurements in figure 6. The final chassis can be seen in figure 7 alongside the whole robot.

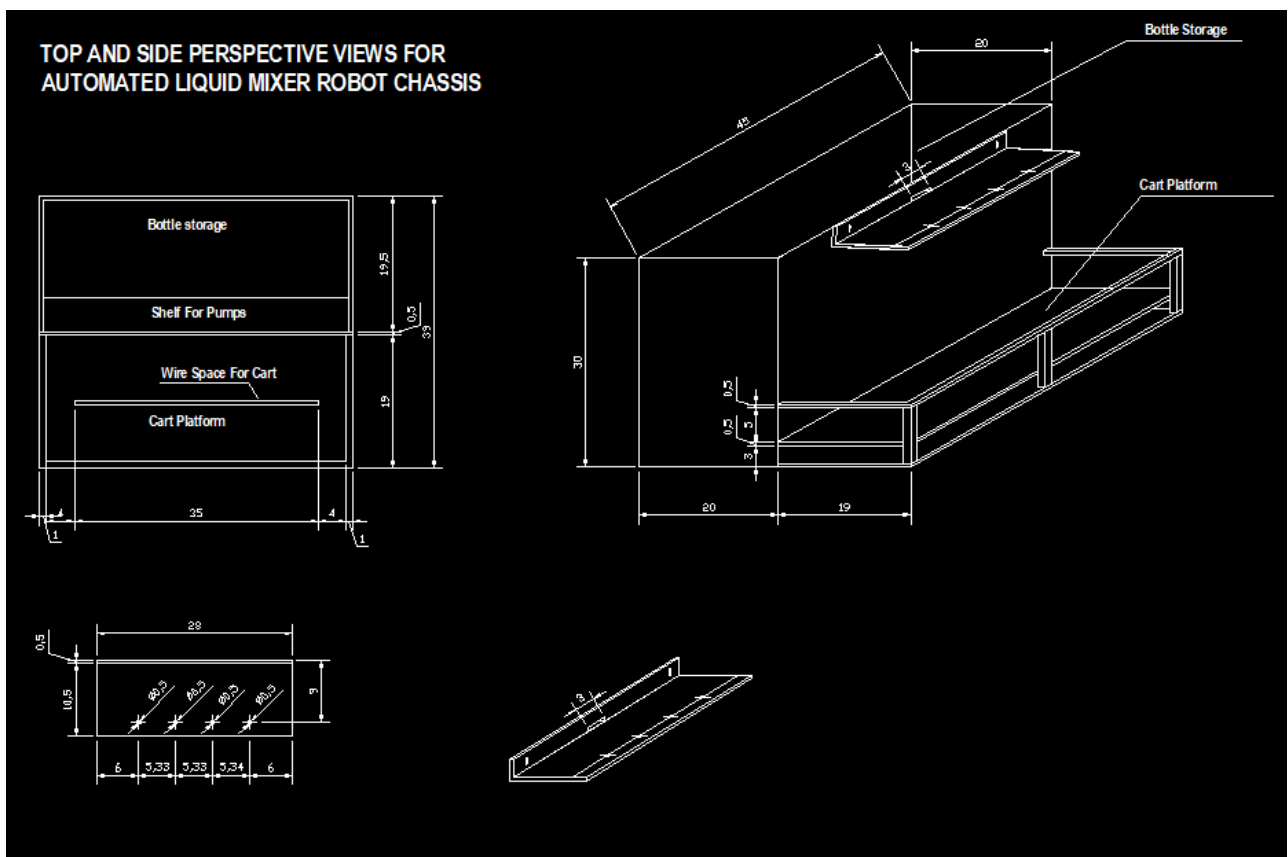


Figure 6 : Chassis Measurements

3. Analysis and Results

3.1 Overview

The main project outcome was that the whole system was built and working. In the demo day we prepared many mixtures and did not have any significant issues. The only issue which we will talk about in limitations is the battery problem (batteries depleting after a couple of hours of usage and cheap batteries depleting much quicker).

3.2 Results

We saw during our tests (especially during demo day which was a stress-testing opportunity) that our robot and mobile application worked well and worked as fully intended when the batteries were full or near full.

The robot can pour specific amounts of specified liquids and not pour any of the liquids the user does not want. The application will only allow mixtures below the maximum specified millilitre amount.

3.3 Limitations

The main limitation of our project was the fact that batteries were a depletable resource and in our demo day after stress testing our project we saw that the batteries were sometimes depleted enough to not allow the robot to function fully. This can be overcome by using SMPS instead of batteries.

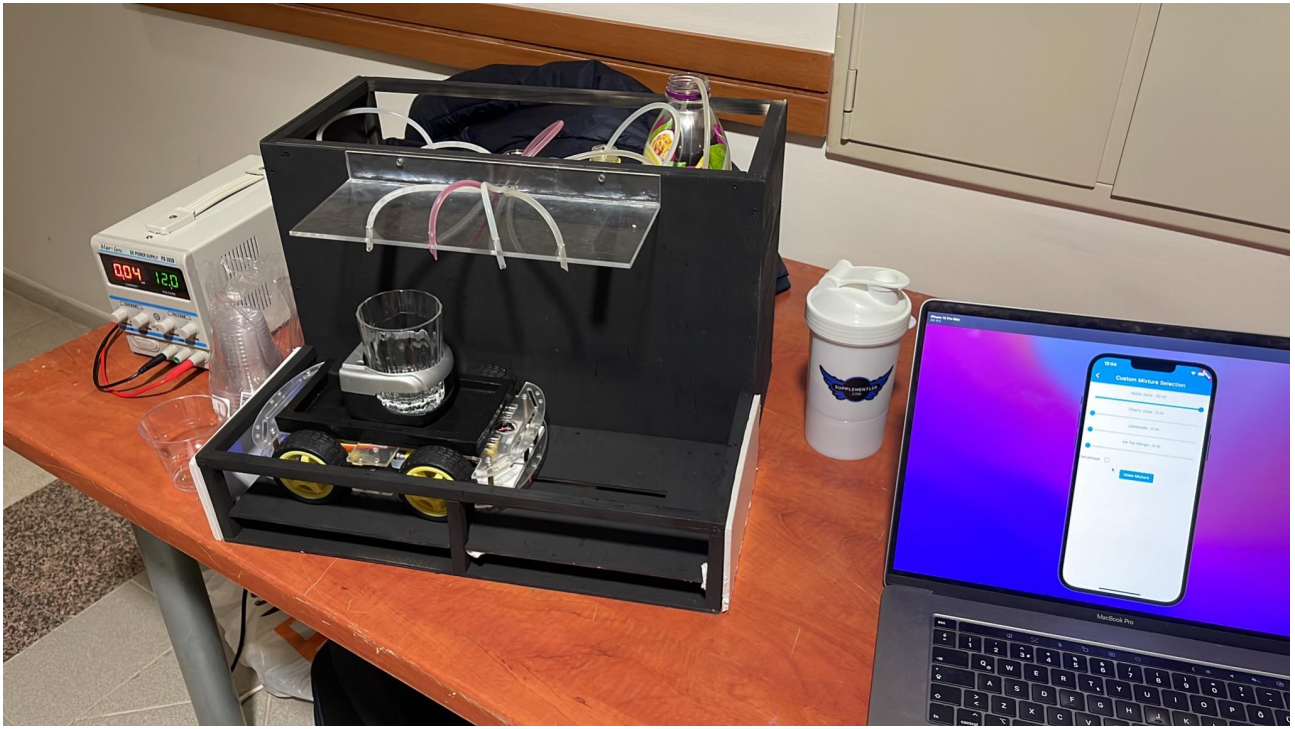


Figure 6: The entire project shown from demo

4. Conclusions

This was a fun project for us. We developed a fully working project that had complicated hardware components and software components. This robot utilised 3 microcontrollers, which meant we had to write 3 different codes that enabled them to work in a coordinated manner. Also, we had a mobile application developed with dart and flutter but we did not know any dart or flutter and learned for this project only. We are satisfied with the final state of this project and that we have such a well functioning end product.

5. References

[1] seafox c, “Arduino cocktail maker | Cheap Barbot”, (Jun. 7, 2020). Accessed: Oct. 16, 2022.

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[3] “Barsys 2.0+”. [Online]. Available: <https://thebarsys.com/products/barsys-2-0> (Accessed Oct. 16, 2022).

[4] “Using Arduino IDE | Adafruit HUZZAH ESP8266 breakout | Adafruit Learning System”.

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[5] “Flutter Documentation | Flutter”. [Online]. Available: <https://docs.flutter.dev/> (Accessed Oct. 16, 2022).

6. Appendixes

Github link for our code: <https://github.com/alikaratas17/Liquid-Mixing-Robot-Project>